

Grade 11 – Term 1

C8 Practice Activity with Solutions

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Evaluated criterion

C8. Uses Student's t -distribution tables, degrees of freedom, and one-tailed or two-tailed areas to determine critical values, central probabilities, and tail probabilities or probability ranges.

1 Introduction to Student's t -Distribution

Problems:

- 1.1. Shape, notation, and degrees of freedom
 - 1.2. Tail areas, symmetry, and critical values
 - 1.3. Reading the Student's t -table
 - 1.4. Worked table examples
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1.1 Shape, notation, and degrees of freedom

Student's t -distribution is a family of continuous probability distributions used to model standardized sample means when the population standard deviation is unknown and is replaced by the sample standard deviation. Write

$$T \sim t_{\nu},$$

where ν (the Greek letter nu) is the number of **degrees of freedom**. Each curve is bell-shaped and symmetric about zero, with total area 1. Compared with the standard normal distribution, a t -distribution has heavier tails: large positive and negative standardized values are more likely.

For an independent random sample of n observations from a normal population, the standardized sample mean

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Here $\bar{X}$ is the sample mean, μ the population mean, and S the sample standard deviation. Estimating the mean leaves $n - 1$ independent deviations, so $\nu = n - 1$. This rule applies to the sample-mean models in this activity; degrees of freedom need not equal $n - 1$ in every statistical model. No calculation of T from raw data is required here.

Small degrees of freedom give heavier tails and larger positive critical values for the same upper-tail area. As ν increases, the curve approaches the standard normal curve. A t -value is dimensionless; it is not an expenditure, return, or delivery time in the original units.



1.2 Tail areas, symmetry, and critical values

For $t > 0$, let $q = P(T > t)$. Symmetry and complementary areas give

$$\begin{aligned} P(T < -t) &= P(T > t) = q, \\ P(T < t) &= 1 - q, \\ P(|T| > t) &= P(T < -t) + P(T > t) = 2q, \\ P(-t < T < t) &= 1 - 2q. \end{aligned}$$

Thus a **one-tailed** probability refers to one end of the distribution. A **two-tailed** probability combines both ends outside $-t$ and t . A **central** probability is the area between these two cutoffs. For any real a , symmetry gives $P(T < a) = P(T > -a)$. Since the distribution is continuous, including or excluding an endpoint does not change its probability.

A **critical t -value** is a cutoff that leaves a specified area in a tail or between symmetric cutoffs. In this document, $t_{\nu,q} > 0$ denotes the cutoff satisfying $P(T > t_{\nu,q}) = q$:

Required area	Cutoff or cutoffs
$P(T > t) = \alpha$	$t = t_{\nu,\alpha}$
$P(T < c) = \alpha \quad (\alpha < 0.5)$	$c = -t_{\nu,\alpha}$
$P(T > t) = \alpha$	$-t_{\nu,\alpha/2}, t_{\nu,\alpha/2}$
$P(-t < T < t) = p$	$-t_{\nu,(1-p)/2}, t_{\nu,(1-p)/2}$



1.3 Reading the Student's t -table

Each entry below is a positive $t_{\nu,q}$; the column heading q is the **upper-tail probability**, not the central area or the combined two-tail area. First choose the row for ν , then convert the requested area into q , and finally read the entry. The extra header rows show the corresponding two-tail and central areas.

Upper tail q	0.10	0.05	0.025	0.01	0.005
Two tails $2q$	0.20	0.10	0.05	0.02	0.01
Central $1 - 2q$	0.80	0.90	0.95	0.98	0.99
Degrees of freedom ν	Positive critical value $t_{\nu,q}$				
5	1.476	2.015	2.571	3.365	4.032
9	1.383	1.833	2.262	2.821	3.250
11	1.363	1.796	2.201	2.718	3.106
14	1.345	1.761	2.145	2.624	2.977
19	1.328	1.729	2.093	2.539	2.861
24	1.318	1.711	2.064	2.492	2.797
29	1.311	1.699	2.045	2.462	2.756

Entries are rounded to three decimal places. Report critical values to this precision. When a question uses a printed entry as its cutoff, report the associated probability as a table-based approximation. If a cutoff lies strictly between two entries, give a probability interval: larger positive cutoffs leave smaller upper-tail areas. Do not interpolate or claim an exact probability between columns. Every sample size used below has its required row in this table.



1.4 Worked table examples

C8 Example 1 – A positive and a negative cutoff

An analyst models a standardized mean return from $n = 10$ independent observations using $T \sim t_{n-1}$. Find a with $P(T > a) = 0.05$ and b with $P(T < b) = 0.05$.

The degrees of freedom are $\nu = 10 - 1 = 9$. Each request concerns one tail. Row 9, column $q = 0.05$, gives 1.833. Therefore,

$$a \approx 1.833, \quad b \approx -1.833.$$

The negative cutoff follows from $P(T < -1.833) = P(T > 1.833) \approx 0.05$.

C8 Example 2 – Two tails and a central area

A standardized household-expenditure mean uses $n = 15$ observations, so $T \sim t_{14}$. Find symmetric cutoffs enclosing 0.95 of the distribution.

The area outside is $1 - 0.95 = 0.05$, shared equally between the two tails. Use $q = 0.05/2 = 0.025$, not 0.05. Row 14 gives

$$t_{14,0.025} \approx 2.145, \quad P(-2.145 < T < 2.145) \approx 0.95.$$

The cutoffs are approximately -2.145 and 2.145 ; equivalently, $P(|T| > 2.145) \approx 0.05$.

C8 Example 3 – A probability range and symmetry

A standardized delivery-time mean has $n = 20$, so $T \sim t_{19}$. Bound $P(T > 2.30)$, $P(T < -2.30)$, and $P(|T| > 2.30)$.

In row 19, $2.093 < 2.30 < 2.539$. The bounding entries belong to $q = 0.025$ and $q = 0.01$, respectively. Thus

$$0.01 < P(T > 2.30) < 0.025, \quad 0.01 < P(T < -2.30) < 0.025.$$

Doubling the equal tail areas and then taking the complement gives

$$0.02 < P(|T| > 2.30) < 0.05, \quad 0.95 < P(-2.30 < T < 2.30) < 0.98.$$



2 Calculating Probabilities and Critical Values Using a t -Table

For each problem, record the degrees of freedom, classify the requested area, and identify the upper-tail column or the two columns bounding it. All sample-mean models below assume independent random observations from a normal population, with unknown population standard deviation replaced by the sample standard deviation; therefore $T \sim t_{n-1}$. Use only the included table. Solutions follow each problem for checking after an independent attempt.

Problems:

- 2.1. Investment returns – right-tail cutoff
 - 2.2. Household expenditures – left-tail cutoff
 - 2.3. Delivery times – two-tailed cutoffs
 - 2.4. Production costs – central cutoffs
 - 2.5. Wages – reading a printed entry
 - 2.6. Sales – bounding a right-tail probability
 - 2.7. Business revenues – bounding two tails
 - 2.8. Prices – interpreting a negative value
 - 2.9. Small business costs – reading verbal areas
 - 2.10. Payroll records – determining the sample size
 - 2.11. Revenue planning – mixed probability ranges
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2.1 Investment returns – right-tail cutoff

A financial analyst uses $n = 10$ daily investment returns. The standardized sample mean has $T \sim t_9$, with $\nu = 9$.

Questions/Tasks



C8 Find the positive cutoff t such that $P(T > t) = 0.10$.

C8 Solution

This is one right tail. Use row 9 and column $q = 0.10$: $t \approx 1.383$.



2.2 Household expenditures – left-tail cutoff

A survey uses $n = 12$ household expenditures. The standardized mean has $T \sim t_{11}$, with $\nu = 11$.

Questions/Tasks



C8 Find the negative cutoff c such that $P(T < c) = 0.025$.

C8 Solution

By symmetry, use the upper-tail column $q = 0.025$ in row 11, then change the sign: $c \approx -2.201$.



2.3 Delivery times – two-tailed cutoffs

A logistics company uses delivery times from $n = 15$ deliveries to model a standardized sample mean.

Questions/Tasks



C8 Determine ν and find $t > 0$ such that $P(|T| > t) = 0.10$. State both cutoffs and the area in each tail.

C8 Solution

$\nu = 15 - 1 = 14$. The combined two-tail area is 0.10, so each tail has $q = 0.05$. Row 14 gives $t \approx 1.761$; the cutoffs are -1.761 and 1.761 .



2.4 Production costs – central cutoffs

A manufacturer uses production costs from $n = 20$ independent batches to model a standardized mean.

Questions/Tasks



C8 Find symmetric cutoffs $-t$ and t enclosing a central probability of 0.98. State the area outside them.

C8 Solution

$\nu = 19$. The outside area is 0.02, so $q = 0.01$. Row 19 gives cutoffs -2.539 and 2.539 .



2.5 Wages – reading a printed entry

A business models a standardized mean wage using a sample of $n = 25$ employees.

Questions/Tasks



C8 Use the table to estimate $P(T > 2.064)$ and $P(T < 2.064)$.

C8 Solution

$\nu = 24$. The entry 2.064 is in column $q = 0.025$. Therefore $P(T > 2.064) \approx 0.025$ and $P(T < 2.064) \approx 0.975$.



2.6 Sales – bounding a right-tail probability

A retailer uses sales from $n = 30$ randomly selected trading days to model a standardized mean.

Questions/Tasks



C8 Give a probability interval for $P(T > 1.90)$, identifying the two table entries used.

C8 Solution

$\nu = 29$. Since $1.699 < 1.90 < 2.045$, the relevant columns are 0.05 and 0.025. Hence $0.025 < P(T > 1.90) < 0.05$.



2.7 Business revenues – bounding two tails

A revenue model uses standardized sample means from $n = 12$ independent businesses.

Questions/Tasks



C8 Give probability intervals for $P(|T| > 2.50)$ and $P(-2.50 < T < 2.50)$.

C8 Solution

$\nu = 11$. Since $2.201 < 2.50 < 2.718$, $0.01 < P(T > 2.50) < 0.025$. Doubling gives $0.02 < P(|T| > 2.50) < 0.05$. The central area therefore satisfies $0.95 < P(-2.50 < T < 2.50) < 0.98$.



2.8 Prices – interpreting a negative value

A price study models standardized mean prices from samples of $n = 15$ shops.

Questions/Tasks



C8 Bound $P(T < -2.30)$ and $P(T > -2.30)$. Explain the role of symmetry.

C8 Solution

$\nu = 14$. The entries $2.145 < 2.30 < 2.624$ imply $0.01 < P(T > 2.30) < 0.025$. Symmetry gives $0.01 < P(T < -2.30) < 0.025$. Taking the complement gives $0.975 < P(T > -2.30) < 0.99$.



2.9 Small business costs – reading verbal areas

A standardized mean operating-cost model uses $n = 6$ businesses.

Questions/Tasks



C8 Find (a) the positive cutoff exceeded in 5% of model outcomes; (b) the symmetric cutoffs with 5% of outcomes outside them in total; and (c) the symmetric cutoffs enclosing the middle 80%.

C8 Solution

$\nu = 5$. (a) One right tail: $q = 0.05$, so $t \approx 2.015$. (b) Two tails: $q = 0.05/2 = 0.025$, so the cutoffs are ± 2.571 . (c) Central area: $q = (1 - 0.80)/2 = 0.10$, so the cutoffs are ± 1.476 .



2.10 Payroll records – determining the sample size

A wage study selects 5 employees from each of 5 departments. Treat all 25 observations as independent; the standardized mean follows the sample-mean model described above.

Questions/Tasks



C8 Find the negative cutoff below which 1% of model outcomes lie. Then find the symmetric cutoffs enclosing the middle 90%.

C8 Solution

$n = 5 \times 5 = 25$, so $\nu = 24$. For the left tail, use $q = 0.01$ and symmetry: $c \approx -2.492$. For the central area, $q = (1 - 0.90)/2 = 0.05$, giving cutoffs ± 1.711 .



2.11 Revenue planning – mixed probability ranges

A standardized mean-revenue model uses samples of $n = 10$ stores.

Questions/Tasks



C8 For a cutoff magnitude of 3.00, bound (a) the probability below -3.00 ; (b) the probability outside the symmetric cutoffs; and (c) the probability between them. Also estimate $P(-1.833 < T < 2.262)$ from the table.

C8 Solution

$\nu = 9$. Since $2.821 < 3.00 < 3.250$, $0.005 < P(T > 3.00) < 0.01$. Thus (a) $0.005 < P(T < -3.00) < 0.01$; (b) $0.01 < P(|T| > 3.00) < 0.02$; and (c) $0.98 < P(-3.00 < T < 3.00) < 0.99$. For the asymmetric interval, the left excluded tail is approximately 0.05 and the right excluded tail is approximately 0.025. Therefore $P(-1.833 < T < 2.262) \approx 1 - 0.05 - 0.025 = 0.925$.

